

Percussive audio mixing with Wave-U-Nets

Bradley Aldous

School of Electronic Engineering and Computer Science
Queen Mary University of London, UK

Abstract—A crucial step in the music production process is creating a cohesive combination mixture from the separate recorded components of a piece of music, aided by a variety of audio effects and techniques. Audio mixing is a far more creative and varied task than its subsequential post-production process: mastering, for which there are numerous AI-driven applications on the market. This research takes a look at a promising deep learning method for automatic mixing of drums in dry and wet scenarios; the Wave-U-Net. Modifications are made to the convolutional filters of the current Wave-U-Net architecture to increase the number of trainable parameters of the model, with the idea that this will aid in better understanding patterns in the audio. Both the Wave-U-Net and modified Wave-U-Net are trained and tested, with the output mixes being evaluated through subjective listening tests and an objective analysis of the waveforms and spectrograms. It is found that the modified Wave-U-Net architecture outperforms the current standard Wave-U-Net in the automatic dry mixing of drums, rivalling the performance of human engineers. However the performance in wet mixing is hindered by an inability to recreate certain wet audio effects. Potential future directions are also established.



1 INTRODUCTION

The availability of music production software, like DAWs (Digital Audio Workstations), has seen music production become fairly commonplace as a hobby amongst younger generations. The skills needed to play or record instruments vary from those needed in the post-production stages of obtaining a finished track: mixing and mastering. These latter stages require a more tuned ear for subtleties in the audio, thus many aspiring producers struggle to create professional sounding mixes and masters due to a lack of experience.

In music production, once the individual tracks for a song are recorded, the producer or a separate mixing engineer creates an audio mixture that is ready for the mastering process. The process of audio mixing refers to the balancing of multitrack audio to produce a single cohesive audio mixture. A suite of audio effects and manipulation techniques are employed in this process, typically including: gain staging or level balancing (setting the relative level of the elements in the mix), panning (placement of the elements within the stereo field), equalization or EQ (adjustment of the timbre of elements within a mix), compression (application of dynamic range compression), and reverberation (application of artificial reverberation). Once a mix is produced, it is sent into the mastering process, which is designed to enhance the overall sound and create consistency across a record, ensuring the tracks are ready for release. At the bare minimum mastering chains typically include EQ, compression and limiting (a type of dynamic range compression used to increase the average volume of a track).

The field of intelligent music production has seen considerable developments over recent years in automating aspects of the music production process, notably in developing commercial applications that provide automated mastering functionalities like LANDR [1] and BandLab [2], based on [3, 4]. However, in comparison, there is a distinct lack of commercial applications for automating the mixing process because of the relative complexity of the two tasks. There are a general set of rules to be learned by human mix engineers which help in obtaining a professional

sounding mix, however it is also crucial to treat each song individually in order to create the optimal mix. This special treatment is defined by a vast number of creative decisions made by audio engineers, whereas mastering chains will often contain the same (or a similar) composition of audio effects for songs of the same genre, meaning more of a strict formula can be followed. This more in-depth treatment makes automating the mixing process a far more difficult task compared to automated mastering.

The primary aim of this project is to explore a current state-of-the-art automatic mixing method, with the intention of building on and improving this method, as well as providing an indication as to potential avenues for future research in this field. Overall, the contributions of this paper can be summarised as follows:

- Modifications to the standard Wave-U-Net architecture are made with the motivation that a larger number of trainable parameters will be able to predict complexities in the audio to a better degree.
- A listening test is conducted and found to show that the modified architecture outperforms the standard architecture in both dry and wet mixing scenarios.
- An exploratory analysis of the resulting waveforms and spectrograms is performed, finding similar results to the listening test.
- There were no currently available PyTorch implementations at time of writing for a mixing variant of the Wave-U-Net, so implementations for both architectures have been produced for this paper.

The outline of this paper is as follows. Section 2 presents a review of prior research in the field of audio, focusing down on the three key areas of automatic mixing research: knowledge-based approaches, machine learning approaches and deep learning approaches. The dataset used in this paper is detailed in depth in Section 3, as well as details of the split made for training, validation and testing partitions. Section 4 presents details of the Wave-U-Net architectures and the processes used to train them. Subjective and objective evaluations are performed in Section 5, in the form of a MUSHRA style listening test and a spectrogram

*email: b.aldous@se14.qmul.ac.uk

analysis respectively. Conclusions are drawn from the results and discussed in Section 6, with potential future areas of research identified in Section 7.

2 RELATED WORK

Deep learning has a long history dating back to the 1940s, originating from the development of perceptrons which mimic biological neurons [5], however it has only risen to prominence in research in the last two decades following the application of a fast, greedy algorithm in learning deep belief nets [6, 7]. Numerous fields of research have benefited greatly through the use of neural networks, and whilst computer vision and natural language processing (NLP) get a lot of buzz for their obvious impact on day to day life [8, 9, 10], there are many important uses for deep learning with audio data. Prior to deep learning, audio machine learning applications would rely on more traditional digital signal processing techniques to perform feature extraction, whereas many deep learning architectures learn these features through observing the data that is given.

Many aspects of audio research, outside of audio mixing, have benefited from the application of machine learning methods, like research based on the manipulation of audio signals like noise suppression and speech enhancement [11, 12], and vocal-related fields like speech generation [13] or automatic speech recognition [14], as well as machine listening tasks like automatic music transcription [15, 16]. In comparison to some of these fields, audio mixing research is still fairly niche, and has a much smaller research effort geared towards it, though it has seen a considerable rise in deep learning approaches in the last decade. The body of work attributed to automatic mixing research can generally be divided into two main approaches: knowledge-based systems (KBS) and machine learning (ML) approaches, which can further be broken down into traditional ML approaches and deep learning (DL) approaches. KBS are built on a knowledge base collected from professionals, like a set of rules or guidelines, from which an algorithm is constructed and then applied to a task, whereas ML and DL approaches consist of using statistical methods and algorithms directly on data to create a system that can be applied to a task. The following literature review is divided into these three sections, which are further divided into segments focusing on particular audio effects where they have been researched. Table 1 clearly presents these groupings in the literature for convenience to the reader.

2.1 Knowledge-based approaches to automatic mixing

A large body of work has been geared towards approaches using KBS in automated gain balancing, in [17] a novel system is designed, based on an interactive algorithm whereby a user can rate mixes and thus guide the system to their personal taste, by making minimal or no assumptions. A significant portion of earlier research in the field focused on the application of automated gain balancing systems to a live environment [18, 19, 20, 21], which is of particular difficulty due to the spatial differences between musicians in a live performance. The presented models in these works generally aim to consider the requirements of all listeners in a space whilst optimising the monitor levels accordingly. There has been some research into producing clear speech audio for TV through background ducking [22], though this is minimal in regard to other applications. Systems employing loudness models in measuring the loudness of each track have been explored

in [23, 24, 25], including comparisons of simple energy-based loudness models and more sophisticated psychoacoustic models, whereby it was found that simple energy-based loudness models performed better. An approach to identify source interference, through the calculation of Source to Distortion Ratios (SDRs), to aid in adjusting gains was developed in [26], where it was found that this approach doesn't yield ideal results, but it does allow for the ability to better identify interference levels in recordings. Gain normalization techniques for changing linear systems suitable for both live and studio use are outlined [27, 28], as well as an implementation whereby the gain of a source is adjusted based on unmasking its spectral content from spectrally related channels [29]. The mathematical relationships between the components within a mix are explored in [30], demonstrating how problems in a mix can be overcome through careful consideration of optimisation parameters and strategies.

Methods for automating panning for multitrack mixes tend to follow similar methodologies to implementations for gain level balancing as the concepts are similar in that panning can be seen as separate gains for right and left channels, though these can be made more complex through identifying source placements in a mix. The first efforts in developing an autonomous stereo panner were framed for a live scenario [37], where spectral analysis aided in determining the position of sources in the stereo space. Further research involved the utilisation of spectral decomposition, constraint rules and cross-adaptive algorithms in performing real-time placement of sources in a mix [38], whereby there was no statistical difference found between mixes attained through this method and those produced by a human engineer. An expansion upon this method can be found in [39], where the constructed system is highly versatile and can be used in both live and post-production environments. Furthermore, two more recent methods both focus on the reduction of masking effects between source signals [40, 41], both of these methods achieve higher ratings through subjective listening tests than those that have come before them, attaining standards comparable to professional human mix engineers.

Whilst not being as completely fundamental in obtaining an audio mix when compared to gain level balancing, automated EQ has seen a notable research effort attributed to it. Early attempts to automate EQ of multitrack audio involved the use of cross-adaptive methods [42], with the design intention to reduce set-up time for live and studio environments. Approaches towards tonal balancing through the use of EQ in mixing and mastering processes have been explored [43, 44], with the former employing the Yule-Walker method, whereby an autoregressive model is fit to the input data to match the target EQ curve. Offline and real-time autonomous EQ systems based on masking reduction have been explored [45], in which it was found that the systems reduce both perceived and objective spectral masking, thus improving the quality of a mix.

Some early work in the field of automated compression research focused on noise gates, a compressor style audio processor that attenuates signals registering below a certain threshold volume [49]. In this work an algorithm applied to noisy kick drum recordings (containing bleed from secondary sources) whereby the attack, release, threshold and hold of a noise gate are automatically set is explored, though while the gate parameters are intuitively correct, there was no further evaluation made. Subsequently, a multiple dynamic range compression (DRC) system designed for application in a multitrack mixing scenario was tested against an

Effect/technique	Type of approach		
	Knowledge-based	Machine learning	Deep learning
Gain	[17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30]	[31, 32, 33, 34, 35, 36]	-
Panning	[37, 38, 39, 40, 41]	-	-
EQ	[42, 43, 44, 45]	[46]	[47, 48]
Compression	[49, 50, 51, 52, 53, 54, 55]	-	[56, 57]
Reverb	-	[58, 59, 60, 61]	-
Multiple	[62, 63, 64, 65, 66, 67, 68, 69, 70]	[71]	[72, 73, 74]

TABLE 1: A review of automatic mixing literature, grouped by type of approach and effect/technique.

expert engineer where the results were found to be comparable [50]. Further work was done to automate the choice of parameter values on a compressor based on feature extraction of an input signal [51], leaving just the threshold as a user specified parameter. However this method is further expanded to take into account the threshold parameter [52], where results can compete or outperform those created by semi-professional human engineers. Compression is a critical effect in a typical mastering chain as it aids in achieving a greater desired loudness for a track, and as such a statistical approach geared towards this purpose has been explored in [53]. Following this, a novel method of compression intending to take the individual listener’s environment into account is devised in [54], whereby noise levels taken through the microphone of a listener’s mobile device affect the controls of the compressor. More recently, work has been done to construct a system that can adaptively change compression timing parameters based on an incoming audio signal with the aim of emphasising transients in drum tracks [55], though as of yet there has been no attempt at an in-depth evaluation of this method.

A large body of work has been aimed at developing comprehensive systems involving more than one of the above effects or techniques. A hugely detailed exploration of almost all aspects of automatic mixing is contained in [62], including in-depth breakdowns of each effect used in the mixing process. An early attempt at automating gain levels, panning and EQ based on probabilistic graphical models can be found in [63], where the implementation was found to perform significantly better than other data modelling tools at the time. Further exploratory analysis has been done on the efficacy of a model using these three effects based on the identification of instruments present in a mix [64]. Autonomous mixing systems based on semantic mixing rules gleaned from mixing engineering literature have been researched [65, 66, 67], with listening tests suggesting a performance equal to that of human engineers. Sound source separation techniques have been used as a preprocessing step of the mixing process in a framework designed for automatic mixing of early jazz recordings [68], where the desired improvements were found to be attainable through a block based framework (which includes panning and EQ). A novel method describing a parameter space containing all possible mixes based on gain, panning and EQ is described in [69], where flaws were found in tempo-estimation algorithms, as such improvements need to be made. Most recently, a study into the minimisation of masking in audio mixing using a new multitrack masking metric alongside the use of subgrouping has been conducted in [70], where it was found that both investigated

methods can be used to improve the perceived quality and clarity of a mix.

The overwhelming majority of early automatic mixing approaches were based on KBS, though in recent years the research effort has shifted more towards ML and DL approaches.

2.2 Machine learning approaches to automatic mixing

A significant portion of ML approaches to automatic mixing are specifically tailored to gain balancing as it is the most fundamental step in obtaining a professional audio mixture. The earliest research into this presents a method whereby the Euclidean distance between spectral histograms of a mix and target sound is calculated and fed into an optimisation algorithm to find the best parameters for that mix [31], where the system was found to accurately and successfully create the mixes. Following this, two works presenting similar methods based on the use of linear dynamical systems can be found in [32, 33], where both systems are found to accurately predict the desired mixing coefficients. A novel method designed to reduce masking effects in gain mixes through the employment of genetic programming has been constructed [34], however, after inspection of the generated mixes, it is discussed that there is significant improvement required for this style of method. Further work using genetic algorithms to optimise an audio mix following user input of a listener is researched [35], where the system is found to be comparable in usability to a traditional fader mixer. A recent method focused on gain mixing of drums [36], where gain parameters are estimated through the least-squares equation and used as target values in various machine learning methods, found that a random forest method significantly outperforms others.

There are very few attempts to strictly automate EQ on an audio track, though very early work using inductive learning through pattern recognition can be found in [46], outperforming existing methods at the time.

However, the automation of reverberation in an ML context has been thoroughly researched in the literature. Work that uses a variety of classifiers on features extracted from audio tracks to control the parameters of the reverberation found that a support vector machine classifier performed best in estimating the parameters [58]. An evolution of this method can be found in [59], yielding similar results. Probabilistic Soft Logic (PSL) rules have been defined and weighted to form a model for use in extracting parameters for reverberation [60], where the resulting mixes didn’t have the desired perceptual qualities, though no data preprocessing was performed which can improve the output. More

recently, a similar method using rules extracted from literature was constructed to allow the real-time control of reverberation parameters based on the incoming signal [61], though no formal evaluation was performed, the authors found some high frequency reverberation to be unpleasant.

An approach to classify the importance of audio features in producing a model for processing raw audio to individual stems has been researched [71], where the selected features are used to train regression models to allow this feature set to reduce further.

2.3 Deep learning approaches to automatic mixing

In recent years deep neural networks (DNNs) have seen a significant growth in their application to various processes in music production, including automatic mixing, often achieving state-of-the-art performance. Some attempts at DL approaches to automating EQ have been made. A novel end-to-end convolutional neural network architecture has been implemented in [47], which approximates the target EQ by processing the audio directly. The model is tested for shelving, peaking, lowpass and highpass EQ, whereby the model was found to perform best in the highpass context. An additional method which takes as input an untreated recording and a target, to predict a set of parameters which can then be used to control an audio effect is designed [48]. This method was first constructed for EQ, though it affords the possibility of expansion to apply to other post-production audio effects.

There have been a few attempts at automating DRC with deep learning approaches, with one notable method focusing on mastering applications [56]. This method uses DNNs to predict DRC and spectral balance enhancement parameters, which are then applied to unmastered audio tracks, however it was found through subjective listening tests that this method could not perform to a human mix standard. Further work done utilising siamese DNN architectures, a class of neural network containing two or more identical subnetworks, to learn feature embeddings characterising the effects of a compressor was performed in [57]. These models were found to outperform previous methods in predicting DRC parameters.

A novel approach that makes use of DNNs for automatic mixing of early jazz recordings, an expansion on a previous method [68], has been presented in [72], for which a brief informal listening test suggested a good performance. Motivated by the limited number of available multitrack audio datasets, a domain-inspired model has been constructed which produces parameters that afford the user the ability to adjust a console of audio effects in order to refine a mix [73]. Through a perceptual evaluation this model was found to outperform baseline methods. Drums are generally one of the first parts of a piece of music to be mixed (dependent on genre) and as such much of the research is geared towards this purpose. A recent deep learning method for performing autonomous drum mixing is adapted in [74], whereby a Wave-U-Net model is trained, and found to perform drum mixing to the standard of a professional mix engineer.

3 DATASET

This work makes use of the ENST drums dataset [75], an audio-visual database designed for use in audio signal processing research. The dataset is comprised of multitrack audio for various drum recordings, including 8 audio stems (each stem corresponding to a separate microphone used in the recording process)

as well as dry and wet mixes of each track. Recordings were taken from sessions with three separate drummers, amounting to roughly 75 minutes of recorded audio for each drummer. The drum kits were built with a basic kick, hi-hat and snare setup with varying numbers of toms and cymbals. Recorded sequences were performed with either sticks, brushes, rods or mallets. There is an additional visual component to this database whereby the drummers were recorded from two different angles, however the video recordings were not of benefit to this study, thus this data was not utilised here.

	Training	Validation	Test	Total
Hits	86	11	11	108
Phrase	111	12	12	135
Solo	7	2	2	11
Minus-One	50	7	7	64
Total	254	32	32	318
Percentage	79.87%	10.06%	10.06%	100%

TABLE 2: The distribution of the dataset following a train, validation, test split of 8:1:1.

The two stereo audio mixes, a dry mix and a wet mix, were made from the 8 audio stems. The dry mixing chain is composed of gain level balancing and panning, without any further processing. In the wet mixing chain, each instrument is additionally processed by an appropriate equalization and compression, as well as a slight reverberation along with dynamic processing (limiter/maximiser).

There were four styles of drumming recorded for this dataset: hits, solo, phrases, and minus-one.

- **Hits** refer to a recording whereby the drummer hits a single piece of their drum kit with a single type of stick.
- **Solo** refers to a freestyle recording whereby the drummer is able to use the entire kit however they like.
- **Phrases** are short recordings performed in a specific style or genre at a range of tempos and complexities.
- **Minus-one** recordings are performances whereby the drummer plays to a rhythm determined by a CD or MIDI file that the drummer is listening to.

The recordings for drummers 1 and 2 in the dataset only consist of 7 microphone recordings (the third tom microphone wasn't used in the recording process), thus, for these multitracks, a zero-padded silent input track is incorporated in the third tom's place. All tracks in the data were used in this research, following a train, validation, test split outlined in detail in Table 2.

4 METHODOLOGY

Presented here is an in-depth description of the two deep learning model architectures, as well as the processes used to train them.

4.1 Wave-U-Net architecture

Recently a model that has gained much interest in the audio research field is the Wave-U-Net model [76] (shortened to WN here), inspired by the U-Net architecture, which was first designed for biological image segmentation [77], and went on to find use in

Modified Wave-U-Net

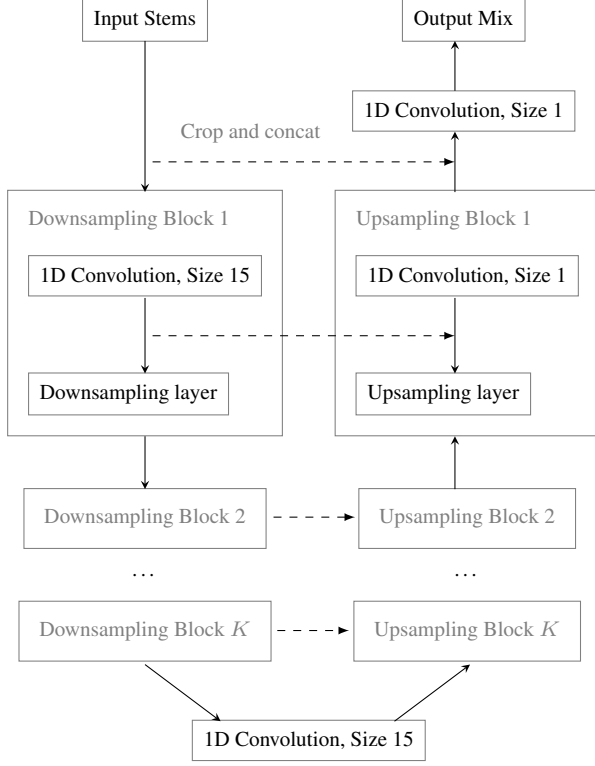


Fig. 1: A block diagram showing the modified Wave-U-Net architecture for K layers, where $K = 10$. Dashed lines represent a crop and concatenation operation. Each convolutional layer is followed by a leaky ReLU activation layer with a negative slope of 0.2, except for the final output convolutional layer of size 1.

similar tasks like binding site prediction of protein structure [78] and image reconstruction [79]. The U-Net class of models are based on encoder-decoder schemes, whereby an image is fed into a series of downsampling blocks comprised of small unpadded convolutional layers and a rectified linear unit (ReLU), and then followed by a small max pooling layer for downsampling. The output of these downsampling layers is then put through a series of upsampling layers, comprised of the same convolutional and ReLU layers as each downsampling block, with a small up-convolutional layer in between to upsample. The U-Net model has seen some applications to the audio field, specifically in audio separation tasks [80, 81] like in [82], where a U-Net architecture is applied to an audio mix in order to split it into the constituent vocal and instrumental components. U-Net has also proven to be effective in the denoising and enhancement of speech [11, 12], as well as the suppression of echo and reverberation [83, 84].

The key difference between the U-Net and WN architectures are as follows. The WN model takes, as input, the raw audio signal, rather than a magnitude spectrogram of the audio signal. Additionally the downsampling and upsampling layers are composed differently; for the U-Net in [82] each downsampling layer is made up of a 5x5 2D convolutional layer, 2D batch normalisation and a leaky ReLU activation layer, whereas in WN, each downsampling layer consists of a 1D convolutional layer of size 15, and a leaky ReLU activation layer. The main benefit of this is that we can avoid the problems related to the reconstruction of the audio signal.

The WN also has a final 1D convolutional layer of size 1 after the last concatenation. WN has proven to be efficient at many of the audio related tasks that U-Net found use in, including audio separation tasks [85], acoustic echo cancellation [86], and speech enhancement [87, 88], as well as mixing tasks [74]. The results in [76, 87] seem to indicate that the WN can achieve a similar or better performance to the U-Net, however a limited dataset means that these conclusions would benefit from further evaluation.

4.2 Modified Wave-U-Net architecture

An additional WN variant was also tested in this research and consisted of the following modifications. The kernel size of the decoder in the model was decreased from 5 kernels to 1 kernels. Furthermore the number of initial filters in the model was increased from 24 to 48. These modifications allow for an increased number of trainable parameters, up to 17,118,882 from 6,183,450, which in turn leads to increased training times. The changes made in this modified Wave-U-Net (MWN) architecture are motivated by the idea that a higher number of trainable parameters can allow for more complex pattern to be identified in the audio. A block diagram of this architecture can be seen in Fig. 1.

4.3 Training procedure

Due to time and hardware limitations, there were some restrictions on aspects of the training process. Firstly, batch sizes up to 16 were tested, where it was found that a batch size of 4 gave the best loss convergence, and so this was chosen for the training.

The models were trained for a total of 150 epochs, with a learning rate scheduler put in place, such that the first 100 epochs ran with a learning rate of 10^{-4} , decreasing to 10^{-5} for the final 50 epochs in a fine-tuning stage. The L1 loss and Adam optimiser were used in the training of these models.

Additionally, input sizes were restricted due to CUDA memory limits, and so input stems 121,843 samples long were used, giving outputs of 89,093 samples in length. This is similar to the methodology applied in [74]. Some experimentation found that initial renditions of the models lacked high frequency content (likely due to the limited length of the training process), so pre-emphasis filters of varying degrees were applied to the inputs before training to account for these shortcomings.

5 EVALUATION

Various methods have been employed to evaluate these methods, the first of which is a subjective perceptual evaluation in the form of a blind listening test. To objectively evaluate the methods, the waveforms and spectrograms of mixes produced by each method have all been visually compared.

5.1 Subjective evaluation

A MUSHRA style listening test was conducted to subjectively evaluate the quality of mixes produced by the two Wave-U-Net models, in turn providing an indication as to the effectiveness of each model in their application to audio mixing.

5.1.1 Setup

The test was conducted online, using the end-to-end listening test platform GoListen [89], following a MUSHRA style test format where a hidden reference and anchor are used to provide an indication of how the systems being tested compare to well-known audio quality levels. A listening test of this style has been shown to be suitable, reliable and accurate in evaluating audio quality [90]. An image of the user interface for the listening test can be seen in Fig. 2.

Participants were first asked to ensure that they were in a quiet environment and were wearing a suitable pair of headphones, such that low frequency data was audible and was taken into account when judging mix quality. Each participant was then presented with a volume adjustment page where they were asked to set the volume of their headphones to a comfortable level given an audio prompt, after which they were asked not to change this level so as to avoid loudness bias.

Following this, each participant was presented with a series of preliminary questions, inspired by the Gold-MSI [91], which were designed to gauge the participant’s level of audio knowledge and critical listening experience, as well as the degree of any hearing impairments they may have.

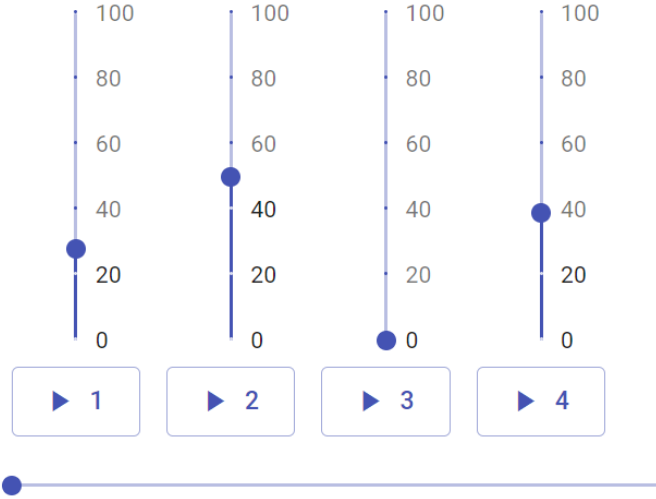


Fig. 2: A screenshot of the listening test user interface on the GoListen platform [89].

5.1.2 Participants

There was a total of 9 participants for the listening test, of which 1 participant noted exhibiting significant hearing impairment symptoms, and thus their response was removed from the analysis. Additionally, one of the wet mix responses for one of the participants was skipped over, so just the remaining nine track responses for this participant were included in the analysis. The test was planned to take no longer than 20 minutes, and as such there were no issues of fatigue for the participants. There were also no financial incentives for anyone involved.

Of the remaining 8 participants, all declared they were over 18 years of age and experienced no hearing impairment symptoms. 3 participants noted that they were experts in the field of audio research, and so a separate analysis has also been done on this

specific group. 5 of the 8 participants reported that they spend at least 1 hour a day critically listening to music.

5.1.3 Composition of listening test

The main body of the test was divided into two sections, dry mix testing and wet mix testing. Each section consisted of 5 questions, each of these questions corresponding to a different track in the test partition of the dataset (10 tracks in total were chosen at random, ignoring ‘hit’ tracks, as little consideration is needed for mixing these). Participants were presented with 4 different audio mixes in each question: the WN output mix, the MWN output mix, a reference track (the dry or wet human engineered mix present in the dataset), and an anchor (constructed in a DAW with randomised gain levels and audio effect parameter values). These tracks were randomised in each question to ensure the reference and anchor are both hidden, as well as ensuring the participant can’t spot patterns across the questions. From these 4 stimuli, participants were asked to score each one from 0-100 based on the perceived quality and balance of the mix, keeping in mind to reflect any differences in the mixes in the scores they give, such as to give a better comparison between the stimuli.

5.1.4 Results

	Mean ranking	Mean expert ranking
Anchor Dry	4.0	4.0
Reference Dry	2.0	2.2
WN Dry	2.2	1.8
MWN Dry	1.8	2.0
Anchor Wet	4.0	3.8
Reference Wet	1.2	1.0
WN Wet	2.6	2.8
MWN Wet	2.2	2.4

TABLE 3: The mean comparative rankings of each type of dry and wet mix for all songs included in the listening test, where Wave-U-Net is denoted WN, and the modified Wave-U-Net is denoted MWN.

To get a better evaluation for the different mixing purposes, the following analysis has been segmented into dry and wet mixing. The mean of the results for each mix was calculated and used to rank the mixes for each song from best to worst (1 to 4, for reference, anchor, WN and MWN). The means of these rankings were then calculated to find the average ranking, across all songs, for each of the four types of mix included in the listening test.

As expected, the anchor performed worst in all cases, however there were some unexpected results regarding the other models. In general, both WN models performed the dry mixing process to the same level as a human engineer. The average ranking of all participants for the dry test reference mixes was 2, however for the standard WN this was 2.2, and for the MWN this was 1.8. This indicates that the modified architecture outperforms both the current WN architecture and a human engineer in a creating a dry mix. However, narrowing this analysis down to just include those participants who reported being experts in the field of audio

research yields some slightly different results. For the expert subcategory, the average ranking for the reference mixes was 2.2, and for the WN was 1.8, and for the MWN was 2, implying that the standard WN outperforms the modified architecture, but both outperform a human engineer. It should be noted here that the expert subcategory is only comprised of three sets of responses, and so the data here is more liable to extreme deviation due to outlying values compared to the analysis of the whole group of participants.

For wet mixing, an informal listening test found that both WN models failed to sufficiently capture the reverberation effects on the reference mix tracks, which has been previously reported [74]. Breaking the listening test down compounds this observation. The average ranking of the wet reference mixes was 1.2, whereas for the WN it was 2.6 and for the MWN it was 2.2, demonstrating that the modified architecture outperforms the traditional architecture in wet drum mixing. This same conclusion can be drawn from the expert subcategory where the rankings are 1, 2.8 and 2.4 respectively. This data is shown in Table 3.

5.2 Objective evaluation

Waveforms and spectrograms have been generated for specific tracks in the test dataset: a disco phrase by drummer 3 for the dry mix, and a blues shuffle phrase by drummer 3 for the wet mix. Drummer 3 tracks have been chosen as the setup for these recordings included a third tom microphone, and phrases have been chosen as they give the largest variety of drum hits in a recording. Both of these choices lead to a greater variety of data to analyse in the resulting waveforms and spectrograms.

Fig. 3 displays the waveforms and spectrograms for the reference, anchor, WN and MWN mixes for both dry and wet mixing scenarios. For the case of dry mixing, it can be seen that comparison of these plots compounds the conclusions drawn from the listening test, in that both the WN and MWN models produce output waveforms of an extremely similar quality to the reference mix. This can also be seen in the corresponding spectrograms. Scrutinising these leads to the conclusion that the MWN produces output mixes that are more similar to the professional mix, than the WN model does, demonstrated by the low-end frequencies in the tail of the track behaving more similarly.

For the wet mixes, it can be seen that the WN and MWN outputs are more alike to each other than either one is to the reference mix. This is explicitly illustrated when observing the space between transients in the spectrograms, as these spaces are relatively empty for the WN and MWN mixes, however there is a lot more response between the transients for the reference mix. This draws the same conclusion as what was drawn from informal listening, in that the reverberation effects in the WN and MWN wet mixes are severely underestimated when compared to the reference mix.

6 CONCLUSION

A modification to the current Wave-U-Net architecture has been proposed and tested, whereby it was found that the performance of both models is as good or better, in both subjective and objective evaluation, than a human engineer’s performance in a dry mixing scenario. However wet mix testing has shown that both models are lacking in their replication of reverberation effects, and are outperformed by a human engineer. Despite this, the modified

Wave-U-Net model was found to marginally outperform the Wave-U-Net model in wet mixing and in most cases of dry mixing, where expert listeners found the Wave-U-Net model to outperform the modified model in dry mixing.

The benefit of deep learning models of this kind is that they have the ability to learn the audio manipulations performed by multiple audio effects more readily and more easily than traditional machine learning approaches. These models learn these manipulations directly through observing the data, and even though the models presented here failed to sufficiently capture the reverberation effects, the subjective and objective evaluation results infer that the models successfully capture the manipulations of the other effects present in the mixing chain. The nature of these models, and the way in which they take in and process data, imply that deep learning approaches are the most promising and interesting candidates for the development of a fully autonomous mixing system.

7 FUTURE WORK

There are certain aspects of this project that could be improved, given fewer limitations. Firstly, experimenting with different hyperparameters, losses and batch sizes could allow for increased performance. Various different losses could be tested to find which function works best for the models included here. Secondly, the training process that each model was run through could be adapted and refined, either by increasing the number of epochs, or using a different learning rate schedule with an early stopping implementation. These changes could all aid in improving the model’s efficiency and accuracy in producing professional standard audio mixtures. Additionally pre-emphasis filters were applied to the inputs to each model in varying degrees to account for any shortcomings in the high frequency output of the models. However it would be good to account for the frequency differences within the architectures themselves, rather than with filters.

Other aspects of this work could’ve been improved if there were fewer limitations. For instance, batch sizes, training times and input data sizes could all have been improved had there not been restrictions enforced on the work due to GPU capabilities.

The evaluation methods could also be expanded as the number of listening test participants was not as great as initially desired, and so a larger study could yield more reliable, and different, results.

Multitrack datasets are very few in number, mainly due to the time to create and compile such a dataset, however if there was a sufficiently large and varied dataset available then the models could be trained to learn the mixing processes of instrumentation beyond drums. Also, due to the nature of multitrack datasets following a strict format, the resultant models are only trained to work on input data of the same size and shape as the dataset. Thus, there is the potential for work to be done to generalise and extend these models to work with any number of input tracks.

Additionally, there could be work to build presets of these models that are trained to produce output mixes possessing certain user-specified qualities. This can potentially be achieved in a number of ways, like through the augmentation of target mixes before they’re fed into the model, or it could be attained as a consequence of modifying the model and its hyperparameters.

The central motivation for this work was inspired by the observation that there seems to be no user-friendly, completely

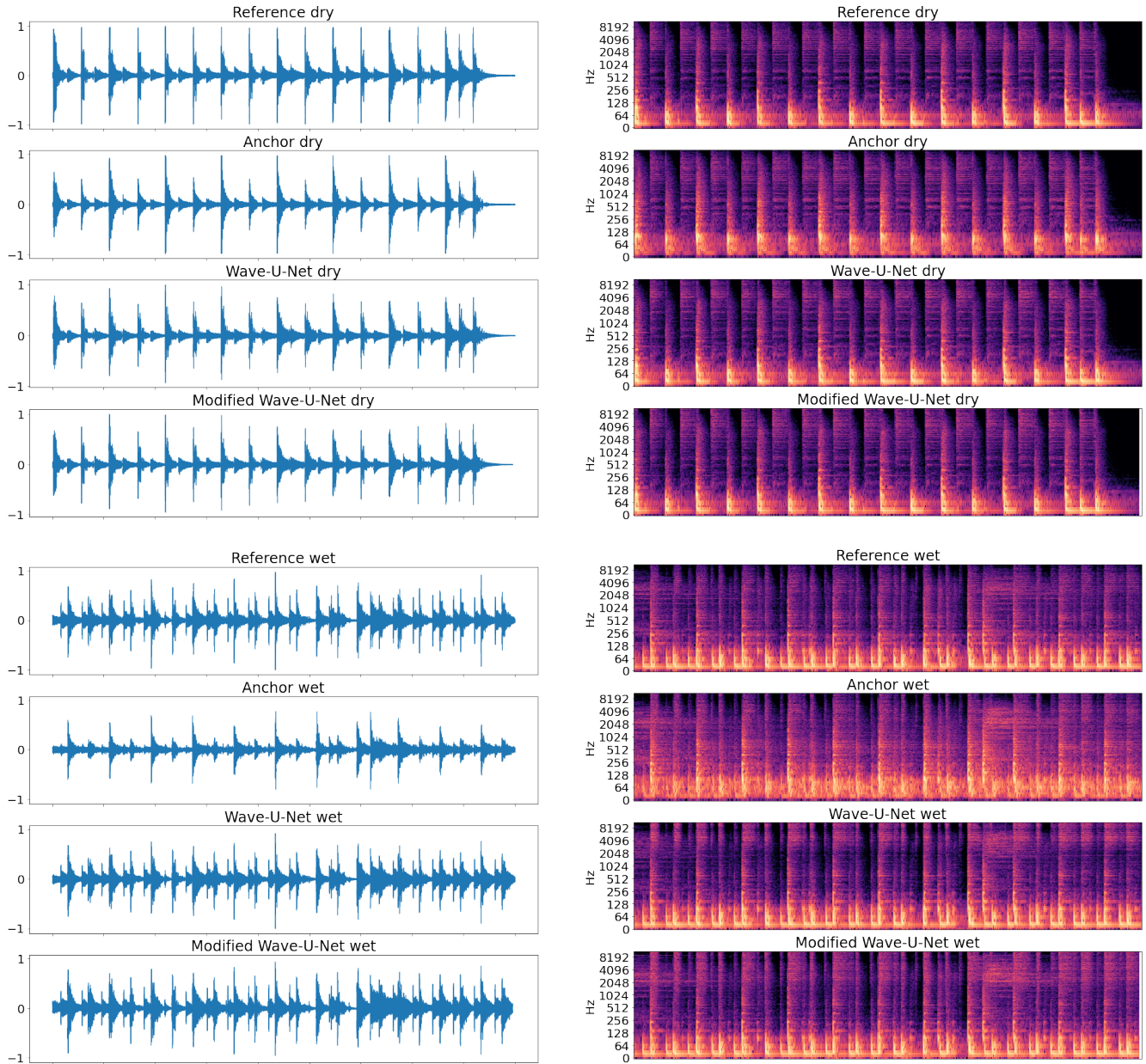


Fig. 3: A plot for comparison of the various mixes outputted by each method. The tracks chosen are track 55 from drummer 3 for the dry mix (a disco phrase), and track 89 from drummer 3 for the wet mix (a shuffle blues phrase). The left side of the plot displays the waveform for each type of wet and dry mix, with the right side displaying the corresponding spectrograms.

autonomous mixing systems available commercially, unlike mastering applications. Much of the above-mentioned possibilities have the potential to compound and culminate in the eventual design of a system of this nature.

ACKNOWLEDGEMENTS

Firstly, I would like to thank Prof Gaël Richard for his assistance in providing the ENST Drums Dataset for use in this project, and Dr Dan Barry for allowing me the use of the GoListen platform for the evaluation.

I would also like to thank Dr Huy Phan and Dr Mathieu Barthet for providing me with access to crucial materials in understanding some of the foundational concepts underlying this research topic.

Additionally, I would like to thank Joseph T. Colonel for giving me his time and assistance in understanding some of the more technical aspects of this project.

Finally, I would like to thank my supervisor, Dr Emmanouil Benetos, for his truly invaluable support and advice throughout every stage of this research, from forming the initial ideas, to the final execution of this project. This work could not have been completed without his help.

REFERENCES

- [1] LANDR: Creative Tools for Musicians. [Online]. Available: <https://www.landr.com/>
- [2] BandLab: Make Music Online. [Online]. Available: <https://www.bandlab.com/?lang=en>

- [3] M. J. Terrell, S. Mansbridge, B. D. Man, and J. D. Reiss, "System and method for performing automatic audio production using semantic data," US Patent 9 304 988, 2016. [Online]. Available: <https://patents.justia.com/patent/9304988>
- [4] J. D. Reiss, S. Mansbridge, A. Clifford, Z. Ma, S. Hafezi, and N. Jillings, "System and method for autonomous multi-track audio processing," US Patent 9 654 869, 2017. [Online]. Available: <https://patents.justia.com/patent/9654869>
- [5] W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *The bulletin of mathematical biophysics* 1943 5:4, vol. 5, pp. 115–133, 12 1943. [Online]. Available: <https://link.springer.com/article/10.1007/BF02478259>
- [6] G. E. Hinton and S. Osindero, "A fast learning algorithm for deep belief nets," *Neural Computation*, vol. 18, pp. 1527–1554, 2006.
- [7] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, pp. 504–507, July 2006.
- [8] T. S. Huang, "Computer vision: Evolution and promise," *CERN European Organization for Nuclear Research Reports-CERN*, pp. 21–26.
- [9] K. Nalbant and Uyanık, "Computer vision in the metaverse," vol. 1, pp. 9–12, December 2021.
- [10] T. Teoh, *Chatbot, Speech, and NLP*, January 2022, pp. 277–287.
- [11] E. Moliner and V. Välimäki, "A two-stage u-net for high-fidelity denoising of historical recordings," February 2022.
- [12] Z. Kang, Z. Huang, and C. Lu, "Speech enhancement using u-net with compressed sensing," *Applied Sciences*, vol. 12, p. 4161, April 2022.
- [13] Y. Chen, "Speech generation by generative adversarial network," *2nd International Conference on Big Data Artificial Intelligence Software Engineering (ICBASE)*, pp. 435–438, September 2021.
- [14] P. Chowdhary, *Automatic Speech Recognition*, April 2020, pp. 651–668.
- [15] E. Benetos, S. Dixon, D. Giannoulis, H. Kirchhoff, and A. Klapuri, "Automatic music transcription: Challenges and future directions," *Journal of Intelligent Information Systems*, vol. 41, pp. 407–434, December 2013.
- [16] A. Klapuri and M. Davy, Eds., *Signal Processing Methods for Music Transcription*. Springer, 2006.
- [17] A. Wilson, "An evolutionary computation approach to intelligent music production informed by experimentally gathered domain knowledge," in *Proceedings of the 2nd AES Workshop on Intelligent Music Production*, 2016.
- [18] M. J. Terrell and J. D. Reiss, "Automatic Monitor Mixing for Live Musical Performance," *J. Audio Eng. Soc.*, vol. 57, pp. 927–936, 2009.
- [19] E. Perez-Gonzalez and J. D. Reiss, "Automatic Gain and Fader Control For Live Mixing," in *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*, 2009.
- [20] M. Terrell and M. Sandler, "An offline, automatic mixing method for live music, incorporating multiple sources, loudspeakers, and room effects," *Computer Music Journal*, vol. 36, pp. 37–54, 2012.
- [21] J. Jiménez-Sauma, "Real-Time Multi-Track Mixing For Live Performance," 2019. [Online]. Available: <https://zenodo.org/record/2550903>
- [22] M. Torcoli, A. Freke-Morin, J. Paulus, C. Simon, and B. Shirley, "Background Ducking to Produce Esthetically Pleasing Audio for TV with Clear Speech," in *146th Convention of the Audio Engineering Society*, 2019.
- [23] S. Fenton, "Automatic Mixing of Multitrack Material Using Modified Loudness Models," in *145th Convention of the Audio Engineering Society*, 2018.
- [24] D. Ward, J. D. Reiss, and C. Athwal, "Multi-track mixing using a model of loudness and partial loudness," in *133rd Convention of the Audio Engineering Society*, 2012.
- [25] G. Wichern, A. S. Wishnick, A. Lukin, and H. Robertson, "Comparison of Loudness Features for Automatic Level Adjustment in Mixing," in *140th Convention of the Audio Engineering Society*, 2015.
- [26] D. Moffat and M. Sandler, "Automatic Mixing Level Balancing Enhanced through Source Interference Identification," in *146th Convention of the Audio Engineering Society*, 2019.
- [27] S. Mansbridge, S. Finn, and J. D. Reiss, "Implementation and Evaluation of Autonomous Multi-track Fader Control," in *132nd Convention of the Audio Engineering Society*, 2012.
- [28] E. Perez-Gonzalez and J. D. Reiss, "An automatic maximum gain normalization technique with applications to audio mixing," in *124th Convention of the Audio Engineering Society*, 2008.
- [29] E. Perez-Gonzalez and J. D. Reiss, "Improved control for selective minimization of masking using inter-channel dependancy effects," in *Proc. of the 11th Int. Conference on Digital Audio Effects (DAFx-08)*, Espoo, Finland, 2008.
- [30] M. Terrell, A. Simpson, and M. Sandler, "The Mathematics of Mixing," *J. Audio Eng. Soc.*, vol. 62, pp. 4–13, 2014.
- [31] B. A. Kolasinski, "A Framework for Automatic Mixing Using Timbral Similarity Measures and Genetic Optimization," in *124th Convention of the Audio Engineering Society*, 2008.
- [32] J. Scott and Y. E. Kim, "Analysis of acoustic features for automated multi-track mixing," in *12th International Society for Music Information Retrieval Conference*, 2011.
- [33] J. Scott, M. Prockup, E. M. Schmidt, and Y. E. Kim, "Automatic multi-track mixing using linear dynamical systems," in *Proceedings of the 8th Sound and Music Computing Conference*, 2011.

- [34] N. Jillings and R. Stables, "Automatic Masking Reduction in Balance Mixes Using Evolutionary Computing," in *143rd Convention of the Audio Engineering Society*, 2017.
- [35] A. Wilson and B. M. Fazenda, "User-Guided Rendering of Audio Objects Using an Interactive Genetic Algorithm," *J. Audio Eng. Soc.*, vol. 67, pp. 522–530, 2019.
- [36] D. Moffat and M. Sandler, "Machine Learning Multitrack Gain Mixing of Drums," in *147th Convention of the Audio Engineering Society*, 2019.
- [37] E. Perez-Gonzalez and J. D. Reiss, "Automatic mixing: live downmixing stereo panner," in *Proc. of the 10th Int. Conference on Digital Audio Effects (DAFx-07), Bordeaux, France*, 2007.
- [38] E. Perez-Gonzalez and J. D. Reiss, "A Real-Time Semi-autonomous Audio Panning System for Music Mixing," *EURASIP Journal on Advances in Signal Processing*, pp. 1–10, 2010.
- [39] S. Mansbridge, S. Finn, and J. D. Reiss, "An Autonomous System for Multi-track Stereo Pan Positioning," in *133rd Convention of the Audio Engineering Society*, 2012.
- [40] P. D. Pestana and J. D. Reiss, "A cross-adaptive dynamic spectral panning technique," in *Proc. of the 17th Int. Conference on Digital Audio Effects (DAFx-14), Erlangen, Germany*, 2014.
- [41] A. Tom, J. Reiss, and P. Depalle, "An automatic mixing system for multitrack spatialization for stereo based on unmasking and best panning practices," in *146th Convention of the Audio Engineering Society*, 2019.
- [42] E. Perez-Gonzalez and J. Reiss, "Automatic equalization of multi-channel audio using cross-adaptive methods," in *127th Convention of the Audio Engineering Society*, 2009.
- [43] Z. Ma, J. D. Reiss, and D. A. A. Black, "Implementation of an intelligent equalization tool using Yule-Walker for music mixing and mastering," in *134th Convention of the Audio Engineering Society*, 2013.
- [44] S. I. Mimitakis, K. Drossos, A. Floros, and D. T. G. Katereolos, "Automated Tonal Balance Enhancement for Audio Mastering Applications," in *134th Convention of the Audio Engineering Society*, 2013.
- [45] S. Hafezi and J. D. Reiss, "Autonomous Multitrack Equalization Based on Masking Reduction," *J. Audio Eng. Soc.*, vol. 63, pp. 312–323, 2015.
- [46] D. Reed, "Perceptual assistant to do sound equalization," in *5th International Conference on Intelligent User Interfaces*, 2000, pp. 212–218.
- [47] M. A. M. Ramírez and J. D. Reiss, "End-to-end equalization with convolutional neural networks," in *Proceedings of the 21st International Conference on Digital Audio Effects (DAFx-18), Aveiro, Portugal*, 2018.
- [48] S. I. Mimitakis, N. J. Bryan, and P. Smaragdis, "One-Shot Parametric Audio Production Style Transfer with Application to Frequency Equalization," in *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing*, 2020.
- [49] M. Terrell, J. D. Reiss, and M. Sandler, "Automatic noise gate settings for drum recordings containing bleed from secondary sources," *EURASIP Journal on Advances in Signal Processing*, pp. 1–9, 2010.
- [50] J. A. Maddams, S. Finn, and J. D. Reiss, "An autonomous method for multi-track dynamic range compression," in *Proc. of the 15th Int. Conference on Digital Audio Effects (DAFx-12), York, UK*, 2012.
- [51] D. Giannoulis, M. Massberg, and J. D. Reiss, "Parameter Automation in a Dynamic Range Compressor," *J. Audio Eng. Soc.*, vol. 61, pp. 716–726, 2013.
- [52] Z. Ma, B. D. Man, P. D. L. Pestana, D. A. A. Black, and J. D. Reiss, "Intelligent Multitrack Dynamic Range Compression," *J. Audio Eng. Soc.*, vol. 63, pp. 412–426, 2015.
- [53] M. Hilsamer and S. Herzog, "A statistical approach to automated offline dynamic processing in the audio mastering process," in *Proc. of the 17th Int. Conference on Digital Audio Effects (DAFx-14), Erlangen, Germany*, 2014.
- [54] A. Mason, N. Jillings, Z. Ma, J. D. Reiss, and F. Melchior, "Adaptive audio reproduction using personal compression," in *AES 57th International Conference, Hollywood, CA, USA*, 2015.
- [55] D. Moffat and M. B. Sandler, "Adaptive Ballistics Control of Dynamic Range Compression for Percussive Tracks," in *145th Convention of the Audio Engineering Society*, 2018.
- [56] S. I. Mimitakis, K. Drossos, T. Virtanen, and G. Schuller, "Deep Neural Networks for Dynamic Range Compression in Mastering Applications," in *140th Convention of the Audio Engineering Society*, 2016.
- [57] D. Sheng and G. Fazekas, "A Feature Learning Siamese Model for Intelligent Control of the Dynamic Range Compressor," 2019. [Online]. Available: <https://doi.org/10.48550/arXiv.1905.01022>
- [58] E. T. Chourdakis and J. D. Reiss, "Automatic Control of a Digital Reverberation Effect using Hybrid Models," in *60th International Conference: DREAMS (Dereverberation and Reverberation of Audio, Music, and Speech)*, 2016.
- [59] E. T. Chourdakis and J. D. Reiss, "A Machine-Learning Approach to Application of Intelligent Artificial Reverberation," *J. Audio Eng. Soc.*, vol. 65, pp. 56–65, 2017.
- [60] A. L. Benito and J. D. Reiss, "Intelligent Multitrack Reverberation Based on Hinge-Loss Markov Random Fields," in *Conference on Semantic Audio*, 2017.
- [61] D. Moffat and M. Sandler, "An Automated Approach to the Application of Reverberation," in *147th Convention of the Audio Engineering Society*, 2019.
- [62] E. Perez-Gonzalez, "Advanced automatic mixing tools for music," Ph.D. dissertation, 2010. [Online]. Available: <https://qmro.qmul.ac.uk/jspui/handle/123456789/614>

- [63] R. Gang, G. Bocko, J. Lundberg, D. Headlam, and M. F. Bocko, "Automatic music production system employing probabilistic expert systems," in *129th Convention of the Audio Engineering Society*, 2010.
- [64] J. Scott and Y. E. Kim, "Instrument identification informed multi-track mixing," in *14th International Society for Music Information Retrieval Conference*, 2013.
- [65] B. D. Man and J. D. Reiss, "A knowledge-engineered autonomous mixing system," in *135th Convention of the Audio Engineering Society*, 2013.
- [66] F. Everardo, "Towards an Automated Multitrack Mixing Tool using Answer Set Programming," in *Proceedings of the 14th Sound and Music Computing Conference, Espoo, Finland*, 2017.
- [67] D. Moffat, F. Thalmann, and M. B. Sandler, "Towards a semantic web representation and application of audio mixing rules," in *Proceedings of the 4th Workshop on Intelligent Music Production, Huddersfield, UK*, 2018.
- [68] D. Matz, E. Cano, and J. Abeßer, "New sonorities for early jazz recordings using sound source separation and automatic mixing tools," in *Proceedings of the 16th ISMIR Conference, M'álaga, Spain*, 2015.
- [69] A. Wilson and B. M. Fazenda, "Populating the Mix Space: Parametric Methods for Generating Multitrack Audio Mixtures," *Applied Sciences*, vol. 7, p. 1329, 2017.
- [70] D. Ronan, Z. Ma, P. M. Namara, H. Gunes, and J. D. Reiss, "Automatic Minimisation of Masking in Multitrack Audio using Subgroups," *ArXiv e-prints*, 2018.
- [71] M. A. M. Ramírez and J. D. Reiss, "Analysis and Prediction of the Audio Feature Space when Mixing Raw Recordings into Individual Stems," in *143rd Convention of the Audio Engineering Society*, 2017.
- [72] S. I. Mimitakis, E. Cano, J. Abeßer, and G. Schuller, "New sonorities for jazz recordings: separation and mixing using deep neural networks," in *Proceedings of the 2nd AES Workshop on Intelligent Music Production, London, UK*, 2016.
- [73] C. J. Steinmetz, J. Pons, S. Pascual, and J. Serrà, "Automatic multitrack mixing with a differentiable mixing console of neural audio effects," in *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*, 2020.
- [74] M. A. Martínez Ramírez, D. Stöller, and D. Moffat, "A Deep Learning Approach to Intelligent Drum Mixing With the Wave-U-Net," *J. Audio Eng. Soc.*, vol. 69, pp. 142–151, 2021.
- [75] O. Gillet and G. Richard, "ENST-Drums: an extensive audio-visual database for drum signals processing," in *7th International Conference on Music Information Retrieval (ISMIR 2006)*, January 2006, pp. 156–159.
- [76] D. Stoller, S. Ewert, and S. Dixon, "Wave-U-Net: A Multi-Scale Neural Network for End-to-End Audio Source Separation," *19th International Society for Music Information Retrieval Conference (ISMIR 2018)*, 2018.
- [77] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," *LNCS*, vol. 9351, pp. 234–241, October 2015.
- [78] F. Nazem, F. Ghasemi, A. Fassihi, and A. M. Dehnavi, "3d u-net: A voxel-based method in binding site prediction of protein structure," *Journal of Bioinformatics and Computational Biology*, vol. 19, April 2021.
- [79] J. Andersson, H. Ahlström, and J. Kullberg, "Separation of water and fat signal in whole-body gradient echo scans using convolutional neural networks," *Magnetic Resonance in Medicine*, vol. 82, pp. 1177–1186, September 2019.
- [80] B.-W. Chen, Y.-M. Hsu, and H.-Y. Lee, "J-net: Randomly weighted u-net for audio source separation," November 2019.
- [81] V. S. Kadandale, J. F. Montesinos, G. Haro, and E. Gómez, "Multi-task U-Net for Music Source Separation," 2020.
- [82] A. Jansson, E. Humphrey, N. Montecchio, R. Bittner, A. Kumar, and T. Weyde, "Singing voice separation with deep U-Net convolutional networks," 2017, pp. 745–751.
- [83] D. León and F. Tobar, "Late reverberation suppression using u-nets," October 2021.
- [84] J. Silva-Rodríguez, M. F. Dolz, M. Ferrer, A. Castelló, V. Naranjo, and G. Pinero, "Acoustic echo cancellation using residual u-nets," September 2021.
- [85] A. Cohen-Hadria, A. Roebel, and G. Peeters, "Improving singing voice separation using deep u-net and wave-u-net with data augmentation," pp. 1–5, September 2019.
- [86] J.-H. Kim and J.-H. Chang, "Attention wave-u-net for acoustic echo cancellation," *Interspeech*, pp. 3969–3973, October 2020.
- [87] C. Macartney and T. Weyde, "Improved speech enhancement with the wave-u-net," November 2018.
- [88] R. Giri, U. Isik, and A. Krishnaswamy, "Attention wave-u-net for speech enhancement," pp. 249–253, October 2019.
- [89] D. Barry, Q. Zhang, P. W. Sun, and A. Hines, "Go listen: An end-to-end online listening test platform," *Journal of Open Research Software*, vol. 9, pp. 1–12, 2021.
- [90] ITU-R, "Bs 1534-1, method for the subjective assessment of intermediate quality level of coding systems," 2001.
- [91] D. Müllensiefen, B. Gingras, J. Musil, and L. Stewart, "The musicality of non-musicians: An index for assessing musical sophistication in the general population," *PLOS ONE*, vol. 9, 2014.